

Task 2 – Sales Performance Auto-Report

Project Summary

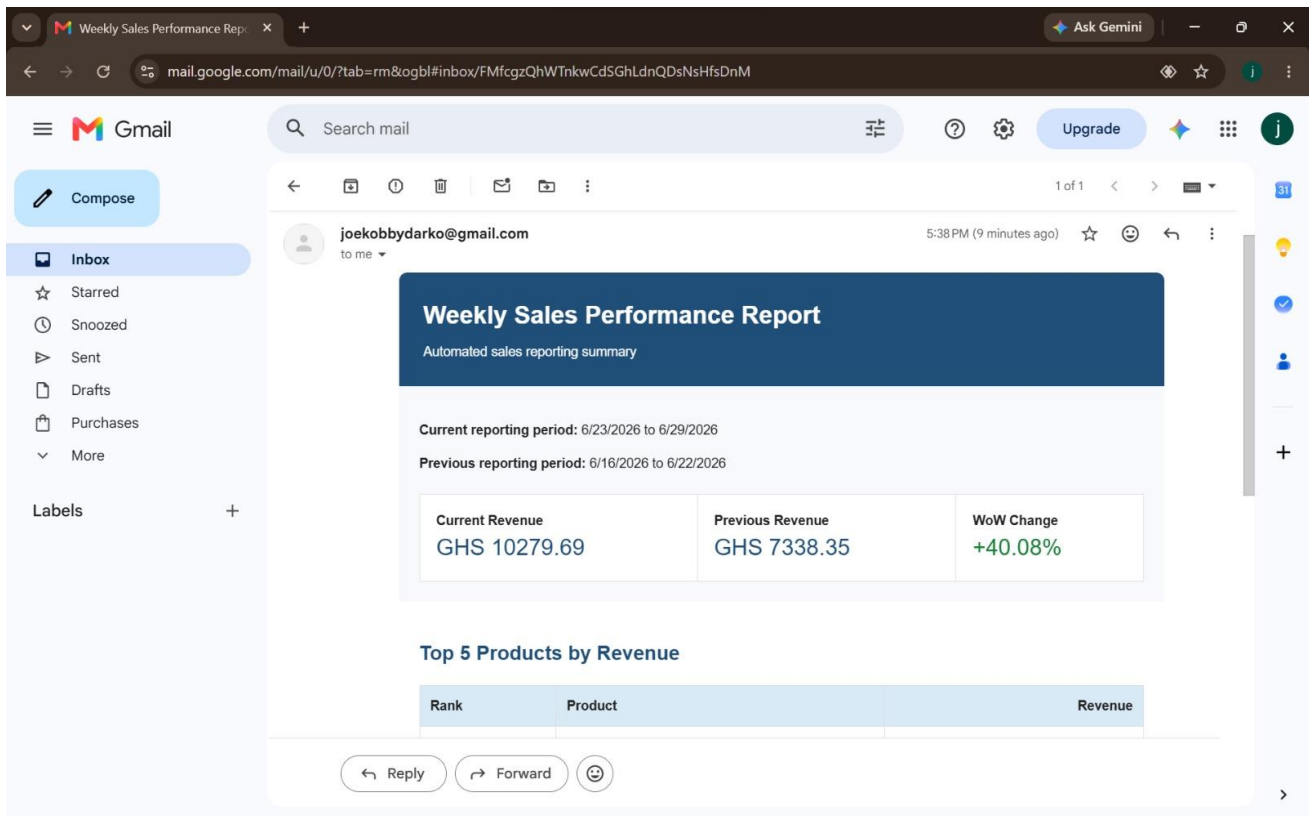
An automated weekly sales reporting system built with Google Apps Script. The script reads a sales CSV from Google Drive, calculates the top 5 products by revenue, revenue by region, and week-over-week percentage change, then sends a professionally formatted HTML email report.

Deliverables Included

- Working Google Apps Script
- Real sample email output screenshot
- Automated reporting logic covering top 5 products, regional revenue, and week-over-week change

Real Sample Output

The screenshot below shows the report successfully delivered by email. The visible report includes the current reporting period, previous reporting period, current revenue, previous revenue, week-over-week change, and the Top 5 Products by Revenue section.



Working Google Apps Script

The following script is the working implementation used to generate and email the weekly sales report.

```
function sendSalesReportEmail() {
  const fileId = '1UWUI6pYiqb7XUMAbOhP01YRXAFsd5F-A';

  const recipientEmail = 'kdarkboadu2005@gmail.com';

  const file = DriveApp.getFileById(fileId);
  const csvContent = file.getBlob().getDataAsString();
  const rows = Utilities.parseCsv(csvContent);

  const headers = rows[0].map(header => header.trim());

  const dateIndex = headers.indexOf('date');
  const productIndex = headers.indexOf('product');
  const totalIndex = headers.indexOf('total');
  const regionIndex = headers.indexOf('region');

  if (
    dateIndex === -1 ||
    productIndex === -1 ||
    totalIndex === -1 ||
    regionIndex === -1
  ) {
    throw new Error(
      'Required columns are missing. Required columns: date, product, total, region'
    );
  }

  const sales = [];

  for (let i = 1; i < rows.length; i++) {
    const row = rows[i];

    if (!row || row.length === 0) {
      continue;
    }

    const dateText = row[dateIndex].trim();
    const product = row[productIndex].trim();
    const region = row[regionIndex].trim();
    const revenue = Number(row[totalIndex]);

    if (!dateText || !product || !region || isNaN(revenue)) {
      continue;
    }

    const saleDate = parseUSDate(dateText);

    if (!saleDate || isNaN(saleDate.getTime())) {
      continue;
    }

    sales.push({
      date: saleDate,
      product: product,
      region: region,
      revenue: revenue
    });
  }

  if (sales.length === 0) {
    throw new Error('No valid sales records were found.');
```

```

previousPeriodEnd.setDate(previousPeriodEnd.getDate() - 1);

const previousPeriodStart = new Date(previousPeriodEnd);
previousPeriodStart.setDate(previousPeriodStart.getDate() - 6);

const currentSales = sales.filter(sale =>
  sale.date >= currentPeriodStart &&
  sale.date <= currentPeriodEnd
);

const previousSales = sales.filter(sale =>
  sale.date >= previousPeriodStart &&
  sale.date <= previousPeriodEnd
);

const currentRevenue = sumRevenue(currentSales);
const previousRevenue = sumRevenue(previousSales);

let weekOverWeekChange = null;

if (previousRevenue !== 0) {
  weekOverWeekChange =
    ((currentRevenue - previousRevenue) / previousRevenue) * 100;
}

const productRevenue = {};

currentSales.forEach(sale => {
  if (!productRevenue[sale.product]) {
    productRevenue[sale.product] = 0;
  }

  productRevenue[sale.product] += sale.revenue;
});

const topProducts = Object.keys(productRevenue)
  .map(product => ({
    product: product,
    revenue: productRevenue[product]
  }))
  .sort((a, b) => b.revenue - a.revenue)
  .slice(0, 5);

const regionRevenue = {};

currentSales.forEach(sale => {
  if (!regionRevenue[sale.region]) {
    regionRevenue[sale.region] = 0;
  }

  regionRevenue[sale.region] += sale.revenue;
});

const revenueByRegion = Object.keys(regionRevenue)
  .map(region => ({
    region: region,
    revenue: regionRevenue[region]
  }))
  .sort((a, b) => b.revenue - a.revenue);

const subject =
  'Weekly Sales Performance Report - ' +
  formatDate(currentPeriodStart) +
  ' to ' +
  formatDate(currentPeriodEnd);

const htmlBody = createEmailBody(
  currentPeriodStart,
  currentPeriodEnd,
  previousPeriodStart,
  previousPeriodEnd,
  currentRevenue,
  previousRevenue,
  weekOverWeekChange,
  topProducts,
  revenueByRegion
);

GmailApp.sendEmail(recipientEmail, subject, 'Your email client does not support HTML.', {
  htmlBody: htmlBody
});

Logger.log('Sales report email sent successfully to: ' + recipientEmail);
}

```

```

function createEmailBody(
  currentPeriodStart,
  currentPeriodEnd,
  previousPeriodStart,
  previousPeriodEnd,
  currentRevenue,
  previousRevenue,
  weekOverWeekChange,
  topProducts,
  revenueByRegion
) {
  const changeText =
    weekOverWeekChange === null
      ? 'N/A'
      : (weekOverWeekChange >= 0 ? '+' : '-') +
        weekOverWeekChange.toFixed(2) +
        '%';

  const topProductsRows = topProducts
    .map((item, index) => {
      return `
        <tr>
          <td style="padding:10px;border:1px solid #ddd;">${index + 1}</td>
          <td style="padding:10px;border:1px solid #ddd;">${item.product}</td>
          <td style="padding:10px;border:1px solid #ddd;text-align:right;">
            GHS ${item.revenue.toFixed(2)}
          </td>
        </tr>
      `;
    })
    .join('');

  const regionRows = revenueByRegion
    .map(item => {
      return `
        <tr>
          <td style="padding:10px;border:1px solid #ddd;">${item.region}</td>
          <td style="padding:10px;border:1px solid #ddd;text-align:right;">
            GHS ${item.revenue.toFixed(2)}
          </td>
        </tr>
      `;
    })
    .join('');

  return `
    <div style="font-family:Arial,sans-serif;max-width:750px;margin:auto;color:#222;">

      <div style="background:#1f4e78;color:white;padding:24px;border-radius:8px 8px 0 0;">
        <h1 style="margin:0;font-size:24px;">
          Weekly Sales Performance Report
        </h1>
        <p style="margin:8px 0 0 0;">
          Automated sales reporting summary
        </p>
      </div>

      <div style="padding:20px;background:#f7f9fc;">
        <p>
          <strong>Current reporting period:</strong>
          ${formatDate(currentPeriodStart)} to ${formatDate(currentPeriodEnd)}
        </p>

        <p>
          <strong>Previous reporting period:</strong>
          ${formatDate(previousPeriodStart)} to ${formatDate(previousPeriodEnd)}
        </p>

        <table style="width:100%;border-collapse:collapse;margin-top:20px;">
          <tr>
            <td style="padding:16px;background:white;border:1px solid #ddd;">
              <strong>Current Revenue</strong><br>
              <span style="font-size:22px;color:#1f4e78;">
                GHS ${currentRevenue.toFixed(2)}
              </span>
            </td>

            <td style="padding:16px;background:white;border:1px solid #ddd;">
              <strong>Previous Revenue</strong><br>
              <span style="font-size:22px;color:#1f4e78;">
                GHS ${previousRevenue.toFixed(2)}
              </span>
            </td>

            <td style="padding:16px;background:white;border:1px solid #ddd;">

```

```

        <strong>Wow Change</strong><br>
        <span style="font-size:22px;color:#188038;">
            ${changeText}
        </span>
    </td>
</tr>
</table>
</div>

<div style="padding:20px;">
    <h2 style="color:#1f4e78;">Top 5 Products by Revenue</h2>

    <table style="width:100%;border-collapse:collapse;">
        <thead>
            <tr style="background:#d9eaf7;">
                <th style="padding:10px;border:1px solid #ddd;text-align:left;">Rank</th>
                <th style="padding:10px;border:1px solid #ddd;text-align:left;">Product</th>
                <th style="padding:10px;border:1px solid #ddd;text-align:right;">Revenue</th>
            </tr>
        </thead>
        <tbody>
            ${topProductsRows}
        </tbody>
    </table>
</div>

<div style="padding:20px;">
    <h2 style="color:#1f4e78;">Revenue by Region</h2>

    <table style="width:100%;border-collapse:collapse;">
        <thead>
            <tr style="background:#d9eaf7;">
                <th style="padding:10px;border:1px solid #ddd;text-align:left;">Region</th>
                <th style="padding:10px;border:1px solid #ddd;text-align:right;">Revenue</th>
            </tr>
        </thead>
        <tbody>
            ${regionRows}
        </tbody>
    </table>
</div>

    <div style="background:#f1f3f4;padding:16px;text-align:center;color:#666;font-size:12px;">
        <p><strong>Sales Performance Auto-Report</strong><br>
Automated weekly sales intelligence report.</p>
    </div>

</div>
`;
}

function parseUSDate(dateText) {
    const parts = dateText.split('/');

    if (parts.length !== 3) {
        return null;
    }

    const month = Number(parts[0]);
    const day = Number(parts[1]);
    const year = Number(parts[2]);

    return new Date(year, month - 1, day);
}

function sumRevenue(sales) {
    return sales.reduce((sum, sale) => sum + sale.revenue, 0);
}

function formatDate(date) {
    return Utilities.formatDate(
        date,
        Session.getScriptTimeZone(),
        'M/d/yyyy'
    );
}

```

